Movie Dataset Visualization Project

Analyzing TMDb Movies Metadata with Python

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Dataset Overview

1

Source

Kaggle – TMDb Movies Dataset

File: movies_metadata.csv

2

Key Attributes

- Title & Release Date
- Genres & Runtime
- Budget & Revenue
- Popularity & Vote Average

3

Analysis Ready

Rich metadata spanning decades of cinema

Perfect for exploring trends and patterns

Data Preprocessing Pipeline

Column Selection

Kept only relevant features for analysis

Date Conversion

Parsed release_date to datetime, extracted year

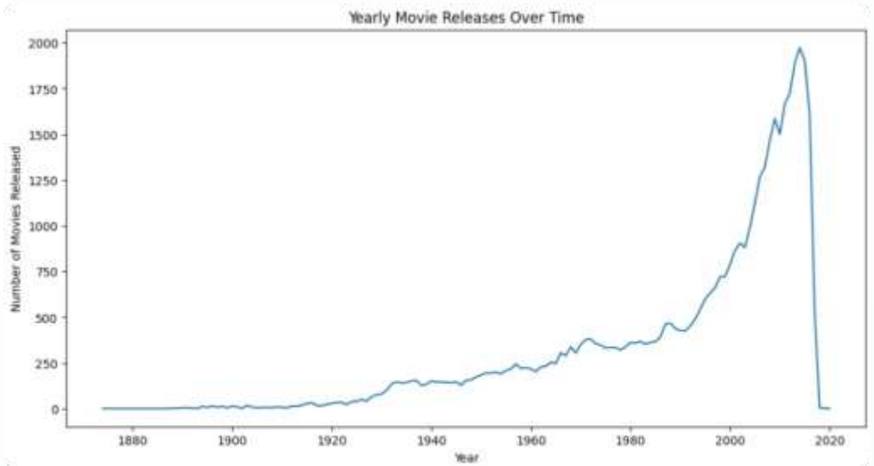
Genre Cleaning

Simplified to primary genre per movie

Data Quality

Removed rows with missing values

Movie Releases Over Time



Line plot reveals exponential growth in movie production. From early cinema to streaming era, releases have surged dramatically.

10x

Growth Rate

Releases increased significantly over past decades

Genre Distribution

Top Genres Dominate

Bar chart shows clear winners in movie production.

1 **Drama**

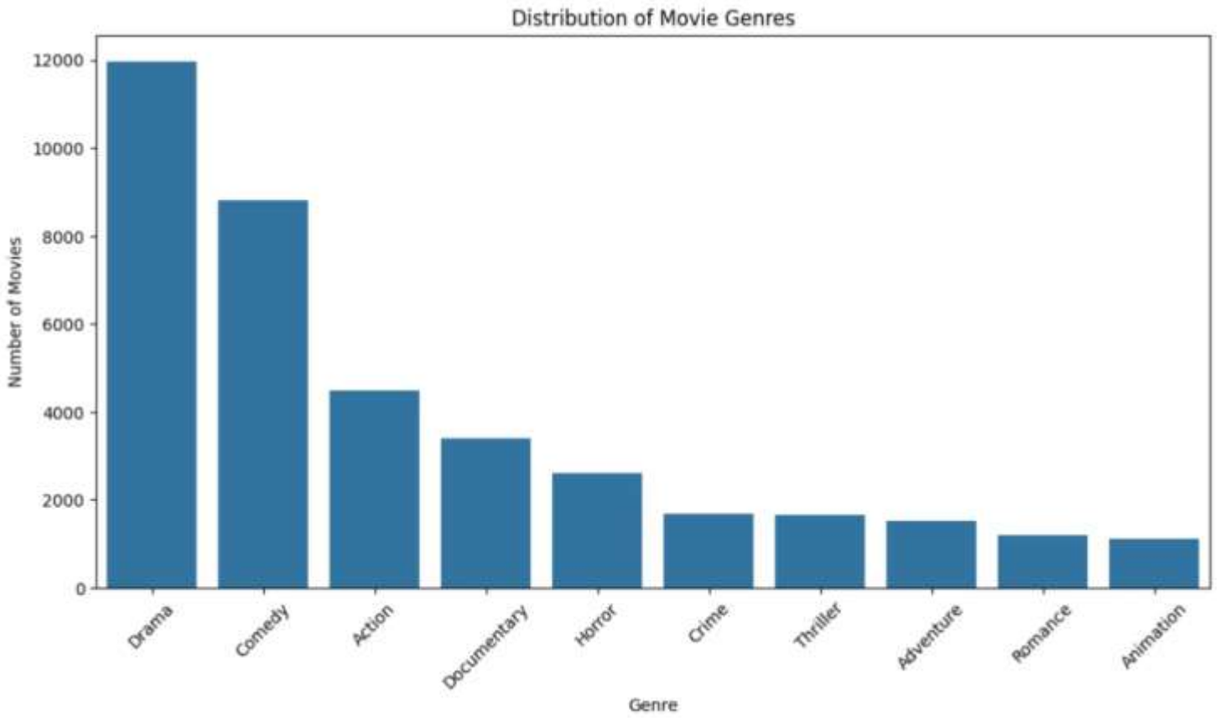
Most produced genre across all time periods

2 **Comedy**

Consistent audience favorite and high output

3 **Action**

Blockbuster appeal drives production volume



Ratings Distribution

Histogram shows most movies cluster around middle ratings—truly exceptional or terrible films are rare.

1 **Average Range**

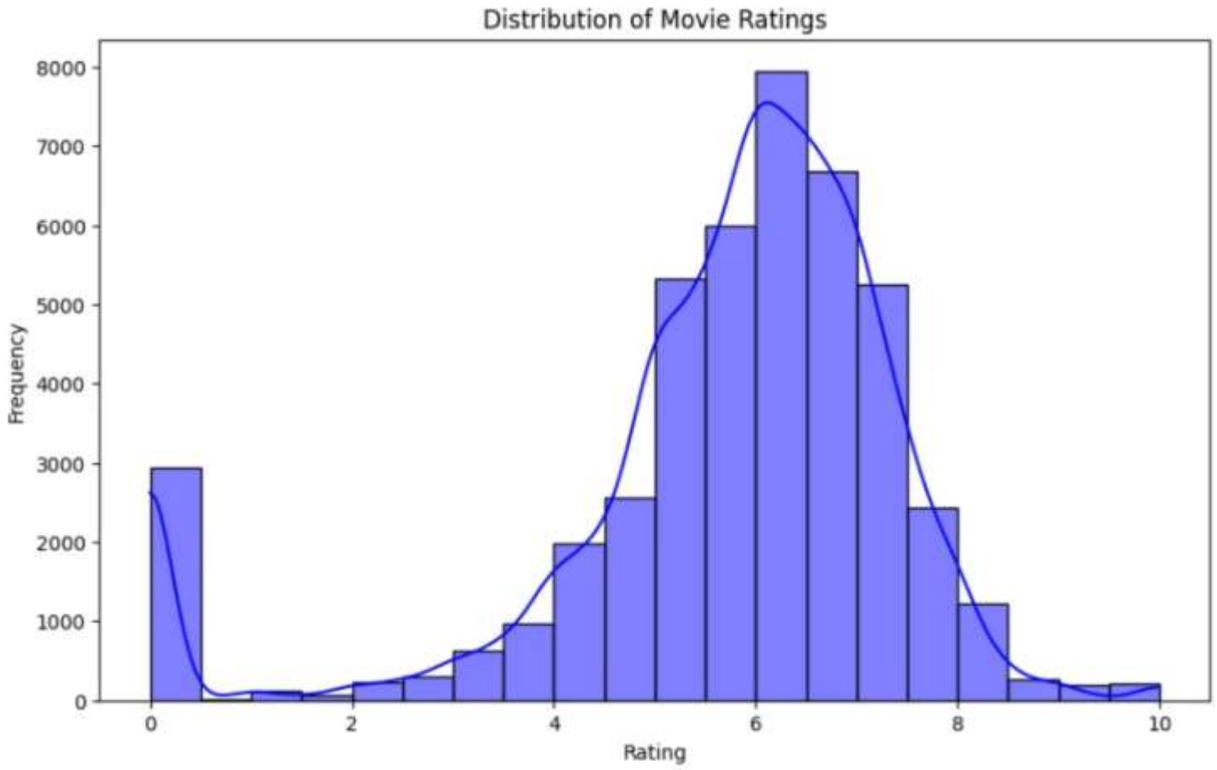
Movies rated between 5 and 7

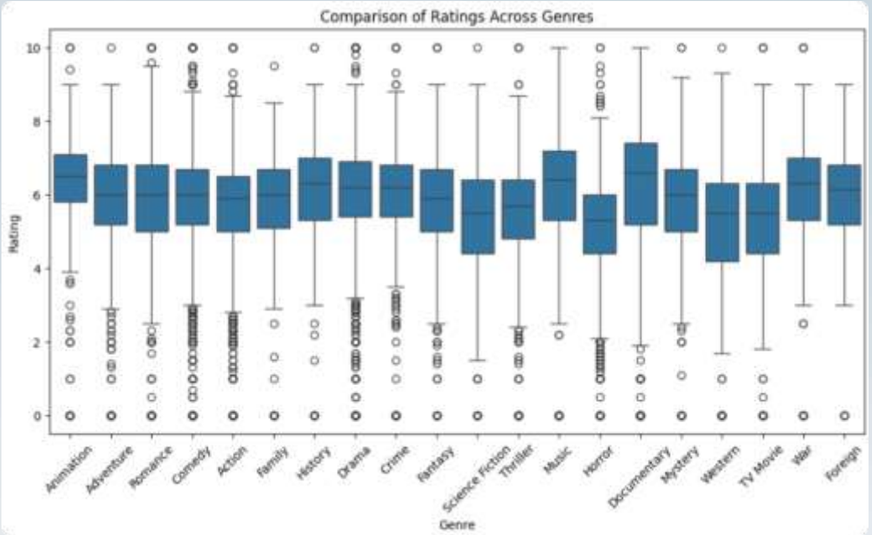
2 **Low Ratings**

Below 5 on the scale

3 **High Ratings**

Above 7 on the scale





Ratings Across Genres

🎬 Documentaries

Consistently score higher—often 6.5+ average

✨ Animation

Strong ratings driven by family appeal and quality standards

🎭 Other Genres

More variable ratings with wider distribution ranges

Boxplot visualization reveals genre quality patterns—some categories maintain higher standards.

Average Ratings per Genre by Decade

Genre Ratings Over Time

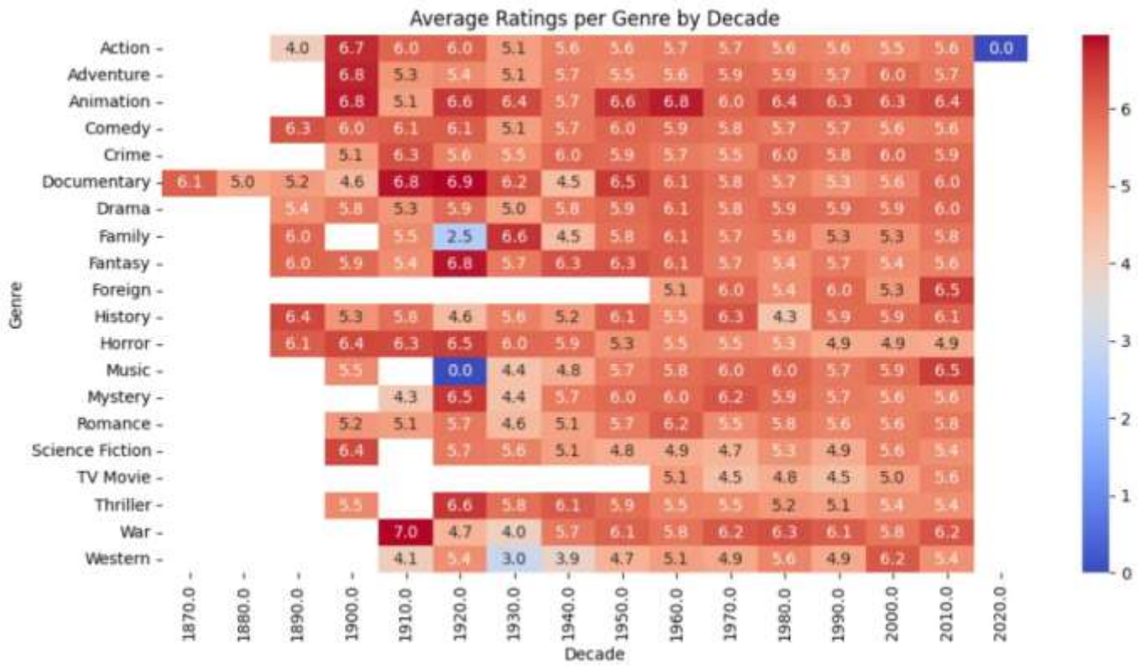
Heatmap visualization reveals how audience ratings vary across decades and genres, highlighting shifts in popularity and quality of genres over time.

Decade-Specific Preferences

Some eras favored Drama & Documentary, while others leaned towards Action & Animation.

Audience & Industry Trends

This helps us understand changing audience tastes and evolving industry directions.



Conclusion:

- Python helps analyze film industry trends effectively.
- Film production has grown over time.
- Drama and Comedy remain popular genres.
- Ratings show the challenges in creativity and competition.
- Visualizations make complex data easy to understand.
- Provides a base for exploring global cinema and audience behavior.